13

The Universe on a Brane

SPECULATIONS ON SPACE AND TIME IN M-THEORY

String theory has one of the most twisted histories of any scientific breakthrough. Even today, more than three decades after its initial articulation, most string practitioners believe we still don't have a comprehensive answer to the rudimentary question, What is string theory? We know a lot about string theory. We know its basic features, we know its key successes, we know the promise it holds, and we know the challenges it faces; we can also use string theory's equations to make detailed calculations of how strings should behave and interact in a wide range of circumstances. But most researchers feel that our current formulation of string theory still lacks the kind of core principle we find at the heart of other major advances. Special relativity has the constancy of the speed of light. General relativity has the equivalence principle. Quantum mechanics has the uncertainty principle. String theorists continue to grope for an analogous principle that would capture the theory's essence as completely.

To a large extent, this deficiency exists because string theory developed piecemeal instead of emerging from a grand, overarching vision. The *goal* of string theory—the unification of all forces and all matter in a quantum mechanical framework—is about as grand as it gets, but the theory's evolution has been distinctly fragmented. After its serendipitous discovery more than three decades ago, string theory has been cobbled together as one group of theorists has uncovered key properties by studying *these* equations, while another group has revealed critical implications by examining *those*.

String theorists can be likened to a primitive tribe excavating a buried spacecraft onto which they've stumbled. By tinkering and fiddling, the tribe would slowly establish aspects of the spacecraft's operation, and